

## The Geospace Event on the Night of 11 August 2000: Results of Ground Geomagnetic Field Measurements From Maitri, Antarctica

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### **Abstract**

Intense bursts of magnetic field variations were recorded on the night of 11 August 2000 by the fluxgate magnetometer installed at the Indian Antarctic Station Maitri (70.8°S, 11.8°E). These variations are ascribed to the southward turnings of the interplanetary magnetic field (IMF) and enhancements in solar wind particle pressure. With an L-value of 4.8, these records track the geospace processes occurring on the night sector of the magnetosphere-ionosphere system.

### **Introduction**

The geographical location of Maitri is unique in that under quiet conditions the station is under the influence of the mid-latitude Sq current system. As the magnetic disturbances grow due to the enhanced solar wind-magnetosphere interaction, Maitri comes increasingly under the auroral oval. The auroral oval expands under such conditions to envelop the sky over Maitri.

Past experiments conducted at the Indian Antarctic Station clearly distinguished the disturbed time behaviour of the geomagnetic field variation from their quiet-time characteristics (Kalra *et al*, 1995). Many important processes occur on the night side of the Earth wherein the geomagnetic field is stretched to form a long 'tail' by the streaming charged particle flow from the Sun. The field lines forming the tail are 'energized' wherever the interplanetary magnetic field turns southward as this favours magnetic field reconnection on the dayside of the Earth. The reconnected field is swept across towards the night side. The magnetic field ultimately releases the energy in the form of sporadic bursts in the tail region that manifest as substorms in the ground geomagnetic data. The channel in

which this energy is routed through is the substorm current wedge that takes a westward path in the high latitude ionosphere. Field-aligned currents constitute the edges of this current pattern (Kisabeth and Rosotoker, 1974).

Controversies still persist as to whether this loading-unloading process is solely responsible for the substorm phenomenon or the directly driven process plays an equally important role in causing the substorm signatures (Rostoker, 1987). The storm-substorm relationship is another area that is gathering increasing attention in recent times.

In this report we have made qualitative assessment of how the geomagnetic field variations as measured on the ground respond to the changing solar wind conditions. Though the results are not new, they provide justification for continuing such studies.

### **Selection of Data**

A fluxgate magnetometer is operated by the Indian Institute of Geomagnetism round the year at Maitri. The magnetometer records daily variations in the X, Y and Z vector components of the geomagnetic field, as well as the pulsations in the same. The data are recorded in both analog (on chart-recorders) and digital (on solid state data loggers) form. Further,, the absorption level of the radio noise at 30 MHz as recorded by the riometer is also monitored continuously. A decrease in the radio intensity at this frequency or an increase in the absorption level would correspond to an increase in the D-region ionization density caused by the precipitation of charged particles along the geomagnetic field lines. These changes would correspond to a field-aligned current that can be detected in the Y-component of the magnetic field.

During the year 2000, the solar activity has reached its peak and many intense solar geophysical events took place then. In this report we focus upon the geomagnetic conditions that prevailed on the night of 11 August 2000. Fig. 1 shows the  $B_x$ ,  $B_y$ , and  $B_z$  components of the interplanetary magnetic field (IMF) in the GSM coordinates measured by the WIND satellite during the period commencing at 1800 UT on 11 August and ending at 1200 UT on 12 August. We show in Fig. 2 the solar wind particle velocities along with the particle density and pressure for the same period, the velocities reached a high of ~650 km/s at 2300 UT and remained at this level till the end of the observation period (1200 UT on 12 August). The ion density and ion pressure showed two peaks at 01.00 UT and 0500 UT on 12 August. The  $B_z$ , component of IMF showed swings to southward values at least on six occasions during the period 1800-0600 UT. The first southward turning occurred at 1930 UT and the last event took place at 0530 UT.

WIND MF>Magnetic Fields Investigation K0>Key Parameter

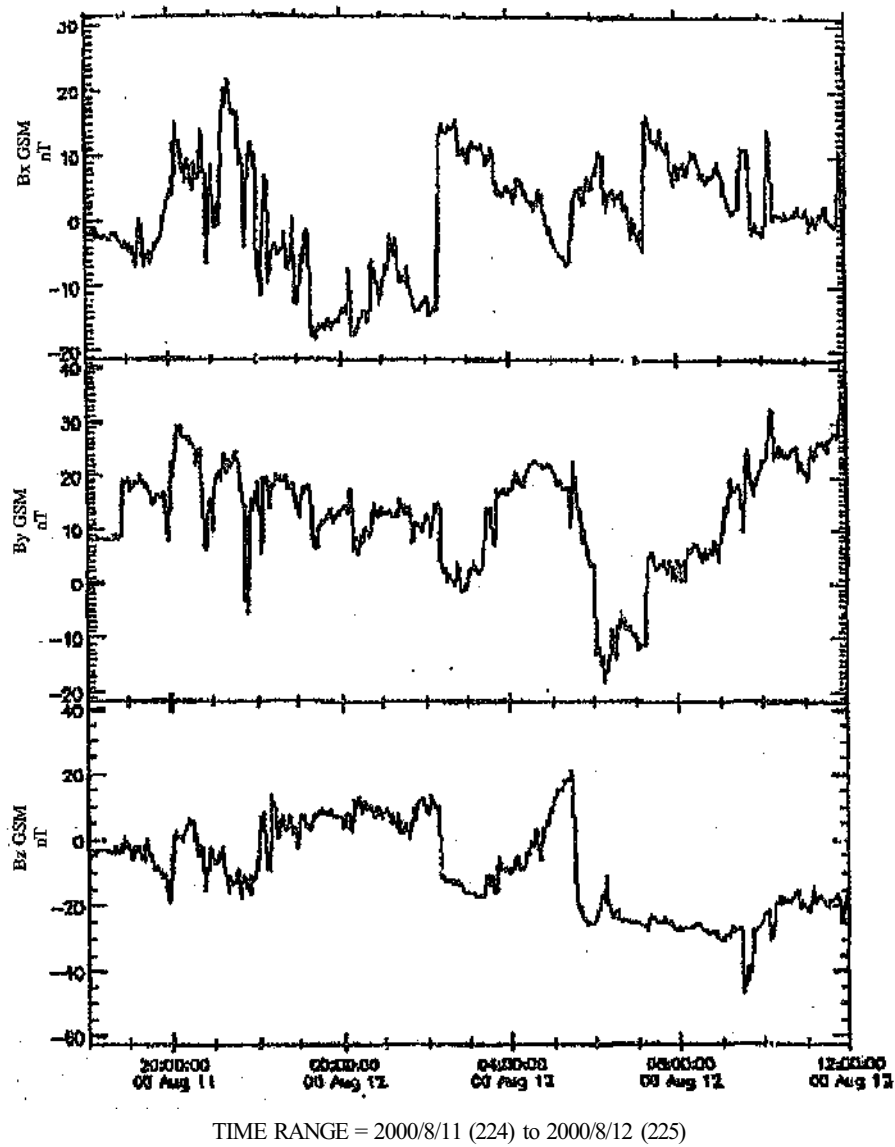


Fig.1:  $B_x$ ,  $B_y$ , and  $B_z$  components of IMF in GSM coordinates as measured by the WIND spacecraft for the period 1800 UT on 11 August 2000 to 1200 UT on 12-August 2000

### Observations

In Fig. 3 the riometer signal and the Z-, X- and Y- components of the geomagnetic field variations recorded at Maitri on the night of 11 August

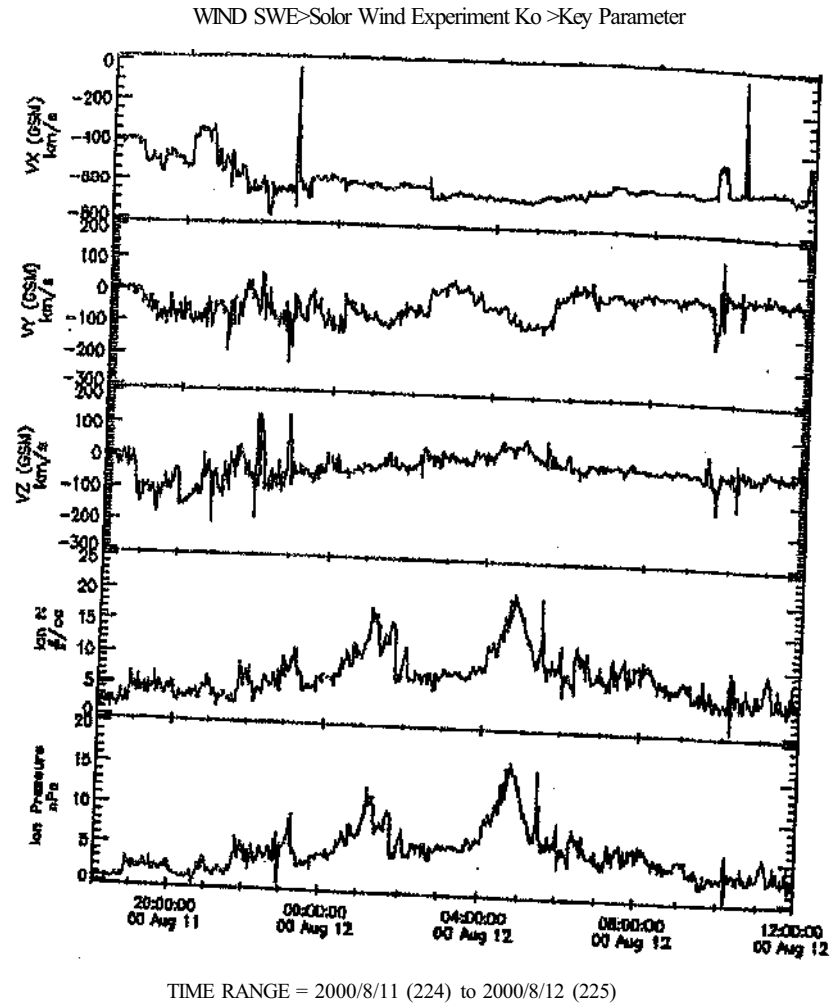


Fig. 2: Same as in Fig. 1 but for solar wind in GSM coordinates, the ion density and ion pressure.

2000 are depicted. In response to the southward turning of the IMF, the ground X-component registered a pronounced northward field during the post-evening hours. This could correspond to the eastward electrojet associated with the directly driven part of a magnetospheric disturbance.

A large negative Z-field implies that the electrojet was located south of

Maitri.

A substorm signature may be noticed in the ground X-component showed a negative depression of  $\sim 400$  corresponding to an intense westward between 2130 and 2200 UT with the onset at 2145 UT. The X-component

electrojet. The Z-component rapidly built up to negative values as large as 300 nT just before 2200 UT and later reached a high positive value in a span of  $\sim 10$  minutes. This clearly indicates an intensification of the westward electrojet and its subsequent movement towards lower latitudes. It may be noticed that the riometer signal gradually decreased to reach lowest values around 2200 UT. Intense particle precipitation might have occurred prior to 2200 UT that can account for this feature. Sharp drop in the Y-component towards negative (westward) values around this time reveals an outward field aligned current path along which the energetic electrons can precipitate. This would also correspond to the westward edge of the substorm current wedge.

It appears that the above event is triggered by the northward turning of the IMF (refer to Fig. 1). Rostoker (1983) has observed many events of expansive phase intensification of the kind reported above during northward turning of IMF. It was highlighted in his work that the largest electrojet perturbations occurred when the IMF  $B_z$  component was northward, strongly suggesting that the energy for the disturbance is derived from the magnetotail rather than directly from the solar wind. It is anticipated that throughout the buildup of the driven system during an episode of southward IMF, energy may be stored in the magnetotail magnetic field. It was argued by Rostoker (1983) that when the IMF turns northward, the magnetotail field will readjust by attempting to return to the configuration it had prior to the southward turning of the IMF. In order to achieve this, the magnetic field strength within the magnetotail will have to reduce. And this energy release might then help in the excitation of a substorm current wedge and a ring current through a rerouting of the high latitude magnetotail current. Another manifestation of the energy release takes place in the auroral ionosphere through the westward electrojet associated with the expansive phase intensification of the substorm and its subsequent Joule dissipation.

An examination of Figs 3 and 4 reveals that the geomagnetic conditions returned to normal around midnight and remained at the quiet-time level till 0300 UT on the morning of 12 August 2000 when a massive substorm expansive phase intensification occurred as reflected in the ground X-component. The IMF turned southward at  $\sim 0230$  UT after a prolonged northward phase (refer to Fig. 1). Intense westward electrojet (the X-component recording more than 1000 nT) associated with this southward turning, perhaps, is linked to a directly driven process. Around this time, the riometer signal shows a drop clearly revealing a particle precipitation event. During the hours following UT, the Y-component shows rapid

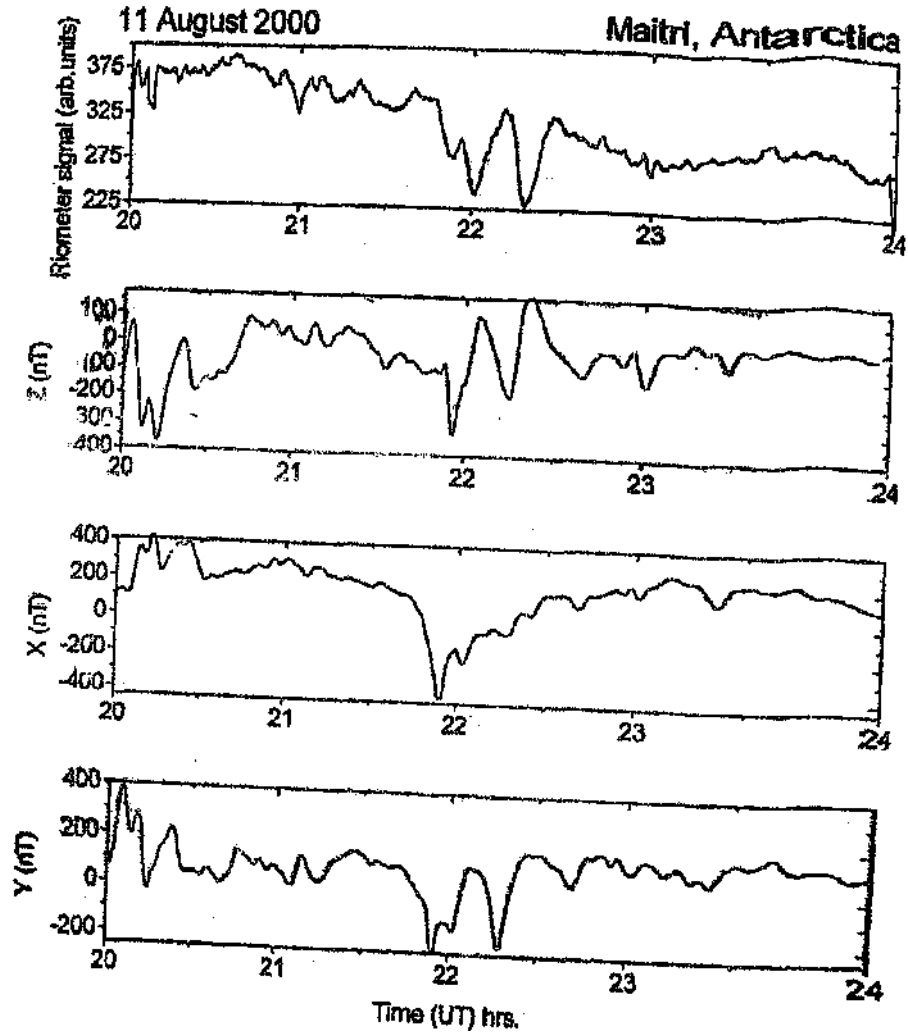


Fig. 3: The X, Y and Z components of the geomagnetic field variations as measured by the fluxgate magnetometer and the riometer signal level at 30 MHz at Maitri, Antarctica on 11 August 2000 in the time interval 2000-2200 UT.

fluctuations implying the bursty nature of the solar wind-magnetosphere interaction and energy injection processes.

After 0500 UT, the conditions appear to return to normal. The IMF gradually becomes northward. At 0530 UT, there is a sharp decrease in the IMF  $B_z$  component and a major substorm has been driven by this southward turning of IMF. At 0600 UT, the riometer signal shows a drop and very

rapid fluctuations in found Y-and components of the geomagnetic field may be noticed around 0700 UT. Similar short-period variations are noticed in the riometer signal.

### **Discussion**

Three substorm signatures were detected on 11/12 August 2000 in the ground geomagnetic data obtained at Maitri, Antarctica. Out of these two were in the early morning hours of 12 August. High solar wind speeds in the range 600-700 km/s were measured by the WIND spacecraft during this period.

This first recorded between 2100 and 2200 UT happened to be an isolated event. Prior to the expansive phase that manifests as westward electrojet (corresponding to a southward field), the IMF was in prolonged southward direction that would have contributed to the buildup of the magnetic flux on the night side and the 'energization' of the tail field. A sudden northward turning appears to have triggered the expansive phase intensification of the substorm current wedge. The Z-component confirms the intensification of the westward electrojet and its subsequence movement towards latitudes north of Maitri. The event is also recorded in the Y-component that clearly indicates the presence of the field-aligned current probably associated with the substorm current wedge. Sharp decrease in the signal and similar temporal variations point toward intense particle precipitation accompanying the field-aligned current.

The geospace was disturbed once again but in the early morning hours of 12 August 2000. A sharp southward turning of IMF around 0200 UT triggers a directly driven process whose manifestation in the ground X-component exceeds field strength of 1000 nT. During the hour following 0400 UT, the IM gradually becomes northward. At 0530 UT the IMF registered a sharp southward turning that is clearly accompanied by the changes in the geomagnetic field as recorded in all components. Between 0600 and 1000 UT, the magnetic disturbances continue unabated revealing a disturbed geospace environment during this period.

The geomagnetic record obtained from Maitri continue to provide valuable information on the processes occurring in the high latitude ionosphere in response to the changing solar wind conditions. Because the field lines originating at Maitri map to distances exceeding four earth radii on the night side, the energy injection effected by the solar wind-magnetosphere interaction and the subsequent energy deposition in the auroral ionosphere can be effectively studied from the magnetometer records such as those obtained from Maitri.

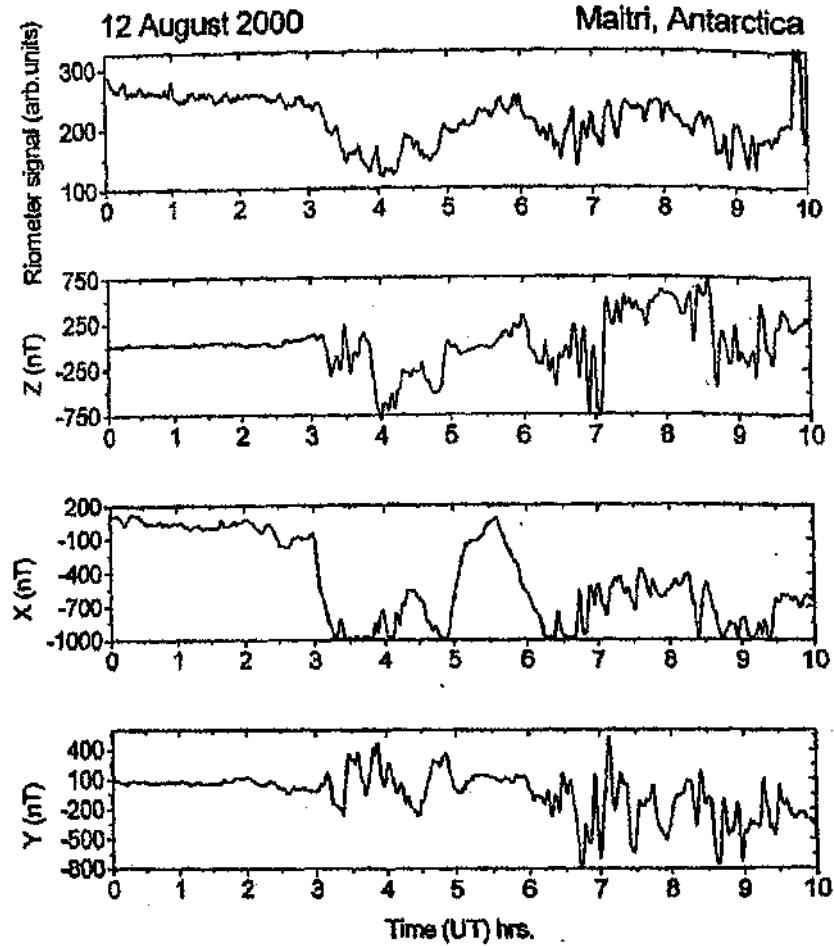


Fig. 4: Same as Fig. 3 but for 12 August 2000 in the time interval 0230-1000 UT.

Future work would address to a few of the issues that are gaining importance in recent times, for example, the directly driven process versus the loading-unloading of magnetic tail energy and the storm-substorm relationship. Data from other stations within Antarctica and those on the conjugate side of the field lines Maitri are proposed to be used in this effort.

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